Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

A loose coil of flexible rope of length L and mass m rests on a frictionless table. A force S is applied at one end of the rope and more and more rope is pulled from the coil with a constant velocity v. A transverse pulse is produced in the rope, while the rope is being pulled from the coil. The velocity of the pulse in the rope (relative to the surface of the table) will be (assuming the pulse moves in the direction of the pull)

(A) zero

(B) $2\mathrm{v}$

(C) $\sqrt{\frac{SL}{m}}$

(D) $\frac{1}{2}\sqrt{\frac{SL}{m}}$

Q.2

A closed and an open pipe, each of $2\ \mathrm{cm}$. diameter are in resonance in fundamental modes with a tuning fork of $700\ \mathrm{Hz}$ when velocity of sound is $336\ \mathrm{m/s}$. The minimum number of such open pipes to be joined end to end and with the closed pipe so that the resulting long closed pipe is in resonance with the same tuning fork is -

(A) 5

(B) 10

(C) 12

(D) 20

Q.3

Sinosudal waves $5.00\mathrm{CM}$ in amplitude is to be be transmitted along a string having a linear mass density equal to $4.00 \times 10^{-2}\ \mathrm{kg/m}$. If the source can deliver a maximum power of $90\ \mathrm{W}$ and the string is under a tension of $100\ \mathrm{N}$, then the highest frequency at which the source can operate it. Is (take $\pi^2 = 10$)

(A) $45.3\ \mathrm{Hz}$

(B) $50\ \mathrm{Hz}$

(C) $30\ \mathrm{Hz}$

(D) $62.3\ \mathrm{Hz}$

Q.4

A wave travels in a medium, from origin towards positive x direction such that kinetic energy of a particle of mass m of the medium varies sinusoidally with an angular frequency ω, its maximum kinetic energy being $\frac{1}{2}\ \mathrm{m}\omega^2\ \mathrm{A}^2$ and at any instant, the shortest distance between two particles at their maximum kinetic energies is λ. Then the equation of the wave is -

(A) $\mathrm{y} = \mathrm{A}\sin\left(2\omega\mathrm{t} - \frac{4\pi}{\lambda}\mathrm{x}\right)$

(B) $y = 2A\cos\left(\frac{\pi}{\lambda}x - \frac{\omega t}{2}\right)$

(C) $\mathrm{y} = \frac{\mathrm{A}}{2}\sin\left(\frac{\omega\mathrm{t}}{2} - \frac{\pi}{\lambda}\mathrm{x}\right)$

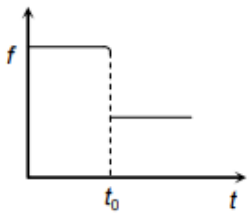
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(D) $y = 2\,A\cos\left(\frac{\pi}{2\lambda}x - 2\omega t\right)$

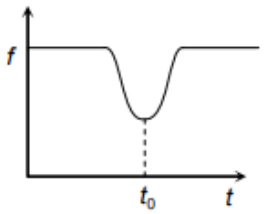
Q.5

A man is standing on a railway platform listening to the whistle of an engine that passes the man at constant speed without stopping. If the engine passes the man at time t_0. How does the frequency f of the whistle as heard by the man changes with time-

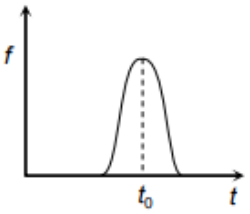
(A)



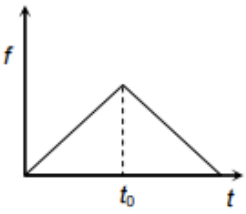
(B)



(C)

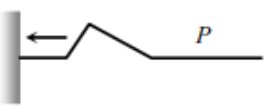


(D)

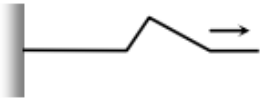


Q.6

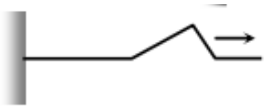
Figure here shows an incident pulse P reflected from a rigid support. Which one of A, B, C, D represents the reflected pulse correctly



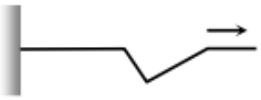
(A)



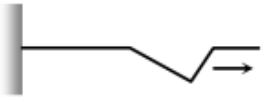
(B)



(C)



(D)



Q.7

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A string of length ' l ' is fixed at both ends. It is vibrating in its 3rd overtone with maximum amplitude 'a'. The amplitude at a distance $l/3$ from one end is

(A) a

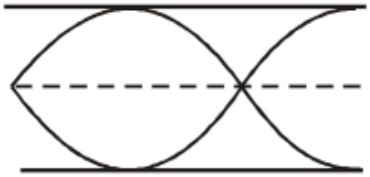
(B) 0

(C) $\frac{\sqrt{3}a}{2}$

(D) $\frac{a}{2}$

Q.8

A longitudinal standing wave is established in a tube open at only one end (see the figure). The frequency of the standing wave is $660\ \mathrm{Hz}$, and the speed of sound in air is $343\ \mathrm{m/s}$. If the length of the tube is $(30 + x)\mathrm{cm}$. Find the value of x.



(A) 2

(B) 4

(C) 6

(D) 9

Q.9

The (x, y) coordinates of the corners of a square plate are $(0, 0), (L, 0), (L, L)$ and $(0, L)$. The edges of the plate are clamped and transverse standing waves are set up in it. If $\mathrm{u(x, y)}$ denotes the displacement of the plate at the point $\mathrm{(x, y)}$ at some instant of time, the possible expression(s) for u is (are) (a = positive constant)

(A) $\mathrm{a}\cos\frac{\pi \mathrm{x}}{2\,\mathrm{L}}\cos\frac{\pi \mathrm{y}}{2\,\mathrm{L}}$

(B) $a\sin\frac{\pi x}{L}\sin\frac{\pi y}{L}$

(C) $a\sin\frac{\pi \mathrm{x}}{\mathrm{L}}\sin\frac{2\pi \mathrm{y}}{\mathrm{L}}$

(D) $\mathrm{acos}\frac{2\pi \mathrm{x}}{\mathrm{L}}\cos\frac{\pi \mathrm{y}}{\mathrm{L}}$

Q.10

Choose incorrect statements related to the Doppler effect (i.e., the shift in frequency of a sound wave) as the temperature increases?

(A) It is greater at higher temperatures, but only in the case of a moving source and a stationary observer.

(B) It is greater at higher temperatures, but only in the case of a moving observer and a stationary source.

(C) It is greater at higher temperatures than at lower temperatures.

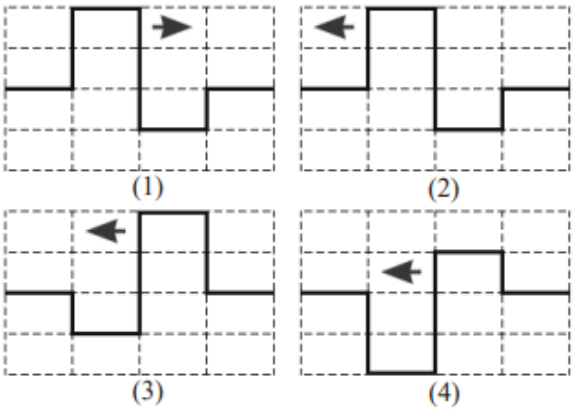
(D) It is less at higher temperatures than at lower temperatures.

Q.11

Pulse 1 in the drawing is moving to the right, while pulses 2,3 , and 4 are moving to the left. Although shown as separated, the four pulses exactly overlap each another at the instant shown. Which combination of these pulses produces a horizontal straight

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line at this instant?



(A) 1

(B) 2

(C) 3

(D) 4

Q.12

Passage:
A vertical pipe open at both ends is partially submerged in water. A tuning fork of unknown frequency is placed near the top of the pipe and made to vibrate. The pipe can be moved up and down and thus length of air column in the pipe can be adjusted. For definite lengths of air column in the pipe, standing waves will be setup as a result of superposition of sound waves travelling in opposite directions. Smallest value of length of air column, for which sound intensity is maximum, is $10\ \mathrm{cm}$. [Take speed of sound, $\mathrm{v} = 344\ \mathrm{m/s}$].
Question: The air column here is closed at one end because the surface of water acts as a wall. Which of the following is correct?

(A) At the closed end of the air column, there is a displacement node and also a pressure node

(B) At the closed end of the air column, there is a displacement node and a pressure antinode

(C) At the closed end of the air column, there is a displacement antinode and a pressure node

(D) At the closed end of the air column, there is a displacement antinode and also a pressure antinode displacement antinode and also a pressure antinode

Q.13

Passage:
A vertical pipe open at both ends is partially submerged in water. A tuning fork of unknown frequency is placed near the top of the pipe and made to vibrate. The pipe can be moved up and down and thus length of air column in the pipe can be adjusted. For definite lengths of air column in the pipe, standing waves will be setup as a result of superposition of sound waves travelling in opposite directions. Smallest value of length of air column, for which sound intensity is maximum, is $10\ \mathrm{cm}$. [Take speed of sound, $\mathrm{v} = 344\ \mathrm{m/s}$].
Question: Frequency of the tuning fork is

(A) $1072\ \mathrm{Hz}$

(B) $940\ \mathrm{Hz}$

(C) $860\ \mathrm{Hz}$

(D) $533\ \mathrm{Hz}$

Q.14

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Passage:
A vertical pipe open at both ends is partially submerged in water. A tuning fork of unknown frequency is placed near the top of the pipe and made to vibrate. The pipe can be moved up and down and thus length of air column in the pipe can be adjusted. For definite lengths of air column in the pipe, standing waves will be setup as a result of superposition of sound waves travelling in opposite directions. Smallest value of length of air column, for which sound intensity is maximum, is $10\ \mathrm{cm}$. [Take speed of sound, $\mathrm{v} = 344\ \mathrm{m/s}$].
Question: Length of air column for second resonance will be

(A) $30\ \mathrm{cm}$

(B) $45\ \mathrm{cm}$

(C) $20\ \mathrm{cm}$

(D) $50\ \mathrm{cm}$

Q.15

Passage:
You are provided with three similar, but slightly different, tuning forks, when A and B are both struck, a beat frequency of $\mathrm{f_{AB}}$ is heard. When A and C are both struck, a beat frequency of $\mathrm{f_{AC}}$ is heard. It was noticed that $\mathrm{f_{AB}} < \mathrm{f_{AC}}$. This experiment is repeated after coating tuning fork A with a little wax. Now it is observed that values of both $\mathrm{f_{AB}}$ and $\mathrm{f_{AC}}$ increase.
Question: Which tuning fork has the highest frequency -

(A) A

(B) B

(C) C

(D) The answer cannot be determined from the information given

Q.16

Passage:
You are provided with three similar, but slightly different, tuning forks, when A and B are both struck, a beat frequency of $\mathrm{f_{AB}}$ is heard. When A and C are both struck, a beat frequency of $\mathrm{f_{AC}}$ is heard. It was noticed that $\mathrm{f_{AB}} < \mathrm{f_{AC}}$. This experiment is repeated after coating tuning fork A with a little wax. Now it is observed that values of both $\mathrm{f_{AB}}$ and $\mathrm{f_{AC}}$ increase.
Question: Which tuning fork has the middle frequency -

(A) A

(B) B

(C) C

(D) The answer cannot be determined from the information given

Q.17

Passage:
You are provided with three similar, but slightly different, tuning forks, when A and B are both struck, a beat frequency of $\mathrm{f_{AB}}$ is heard. When A and C are both struck, a beat frequency of $\mathrm{f_{AC}}$ is heard. It was noticed that $\mathrm{f_{AB}} < \mathrm{f_{AC}}$. This experiment is repeated after coating tuning fork A with a little wax. Now it is observed that values of both $\mathrm{f_{AB}}$ and $\mathrm{f_{AC}}$ increase.
Question: Choose the correct statement -

(A) $(f_B - f_A) < (f_C - f_A)$

(B) $f_C > f_B > f_A$

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(C) $(f_B - f_A) > (f_C - f_A)$

(D) Both (A) and (B)

Q.18

Match the column correctly. (Ratio are wrt fundamental frequency

Column I	Column II
(A) Ratio of consecutive overtones is 2 : 3 : 4	(p) Open organ pipe
(B) Ratio of consecutive overtones is 3 : 5 : 7	(q) String vibration
(C) Ratio of consecutive harmonics 1 : 3 : 5	(r) Closed organ pipe at one end
(D) All even and odd harmonics are present	(s) Sonometer

(A) A-(pqs) B-(q) C-(qr) D-(pqs)

(B) A-(pqs) B-(qr) C-(qr) D-(pqs)

(C) A-(pqs) B-(qr) C-(q) D-(pq)

(D) A-(pqs) B-(qr) C-(qr) D-(pq)

Q.19

A train moves towards a stationary observer with speed $34\ \mathrm{m/s}$. The train sounds a whistle and its frequency registered by the observer is $\mathrm{f_1}$. If the train's speed is reduced to $17\ \mathrm{m/s}$, the frequency registered is $\mathrm{f_2}$. If the speed of sound is $340\ \mathrm{m/s}$ then the ratio $\mathrm{f_1/f_2}$ is $(10 + \mathrm{x})/18$. Find the value of x.

Q.20

Ultrasound in Medicine. A $2.00 - \mathrm{MHz}$ sound wave travels through a pregnant woman's abdomen and is reflected from the fetal heart wall of her unborn baby. The heart wall is moving toward the sound receiver as the heart beats. The reflected sound is then mixed with the transmitted sound, and 72 beats per second are detected. The speed of sound in body tissue is $1500\ \mathrm{m/s}$. Calculate the speed of the fetal heart wall at the instant this measurement is made.

Answers & Solutions

Q.1 Answer:

$\sqrt{\frac{SL}{m}}$

Solution:

Let 'dm' mass adds up in 'dt' time then, tension in the string

$$T = \frac{dp}{dt}$$

$$S = dm = \mu v dt, \mu = \frac{m}{L}$$

$$dm = \frac{m}{L} v dt$$

$$S = v\frac{dm}{dt} = \frac{m}{L}v^2 \Rightarrow v = \sqrt{\frac{SL}{m}}$$

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Relative to surface of table

Q.2 Answer:

20

Solution:

End corrections : Closed $0.3\ \mathrm{d}$

Open : $0.6\ \mathrm{d}$

Let x **be the number of open pipes required.**

Total end correction not available in the long pipe $= 0.3\ \mathrm{d}$ **(for closed pipe)** $+(\mathrm{x}-1) \times 0.6\ \mathrm{d}$ **(for intermediate open pipes)** $+0.3\ \mathrm{d}$ **(for the last open pipe)** $= \mathrm{x} \times 0.6\ \mathrm{d}$

This must equal the one-less half wave $= \frac{\lambda}{2} \Rightarrow \mathrm{x} = \frac{\lambda}{1.2}\ \mathrm{d}; \lambda = \frac{336}{700} = 0.48\ \mathrm{m}\ \mathrm{d} = 0.02\ \mathrm{m} \Rightarrow \mathrm{x} = 20$

Alternate method :

Closed pipe : $\ell_1 + 0.3\ \mathrm{d} = \frac{\lambda}{4} \ldots (1)$

Open pipe : $\ell_2 + 0.6\ \mathrm{d} = \frac{\lambda}{2} \ldots (2)$

Long pipe : $\ell_1 + \mathrm{x}\ell_2 + 0.3\ \mathrm{d} = \frac{\mathrm{n}\lambda}{2} + \frac{\lambda}{4} \ldots (3)$

From the equation,

(1) & (3), $x\ell 2 = \frac{n\lambda}{2} \Rightarrow \frac{n}{x} = \frac{\ell_2}{\lambda/2}$

Substituting (2), $\frac{\mathrm{n}}{\mathrm{x}} = \frac{\frac{\lambda}{2} - 0.6\ \mathrm{d}}{\lambda/2} = 1 - \frac{1.2\ \mathrm{d}}{\lambda} = 1 - \frac{1.2 \times 0.02}{0.48} = 1 - \frac{1}{20} = \frac{19}{20}$

$\Rightarrow \quad \frac{x}{n} = \frac{20}{19}, \min x = 20$

($\therefore$ **both should be integers)**

Q.3 Answer:

$30\ \mathrm{Hz}$

Solution:

$$P = \frac{1}{2}\mu\omega^2 A^2 v, \text{ where}$$

$$v = \sqrt{\frac{T}{\mu}}$$

$$\because f = \omega/2\pi$$

$$\Rightarrow f = \frac{1}{2\pi}\sqrt{\frac{2P}{A^2\sqrt{T\mu}}}$$

Using data,

$\mathrm{f} = 30\ \mathrm{Hz}$

Q.4 Answer:

$y = 2A\cos\left(\frac{\pi}{\lambda}x - \frac{\omega t}{2}\right)$

Solution:

Let $\mathrm{y} = \mathrm{A}' \sin\left(\omega' \mathrm{t} - \frac{2\pi}{\lambda'}\mathrm{x}\right)$ **be the required equation.**

Then angular frequency of KE **oscillation is** $2\omega'$**.**

$\therefore \quad 2\omega' = \omega \Rightarrow \omega' = \frac{\omega}{2} \ldots (1)$

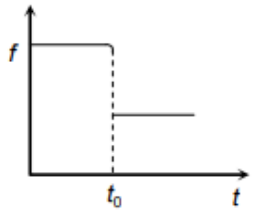
$\mathrm{Max.KE} = \frac{1}{2}\ \mathrm{m}\omega'^2\ \mathrm{A}'^2$

$\Rightarrow \quad \frac{1}{2}\ \mathrm{m}\omega'^2\ \mathrm{A}'^2 = \frac{1}{2}\ \mathrm{m}\omega^2\ \mathrm{A}^2 \Rightarrow \left(\frac{\omega}{2}\right)^2\ \mathrm{A}'^2 = \omega^2\ \mathrm{A}^2 \Rightarrow \mathrm{A}' = 2\ \mathrm{A} \ldots (2)$

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Shortest distance between displacement maximum and minimum $= \frac{\lambda'}{2} \Rightarrow \frac{\lambda'}{2} = \lambda \Rightarrow \lambda' = 2\lambda \therefore \quad y = 2\,A\sin\left(\frac{\omega t}{2} - \frac{\pi x}{\lambda}\right)$ **or it corresponding cos equivalent.**

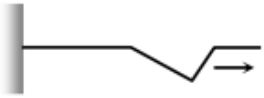
Q.5 Answer:



Solution:

When the train is approaching the stationary observer frequency heard by the observer $n' = \frac{v+v_0}{v}n$.
When the train is moving away from the observer then frequency heard by the observer $n'' = \frac{v-v_0}{v}n$
It is clear that n' **and** n'' **are constant and independent of time.**
Also and $n' > n''$.

Q.6 Answer:



Solution:

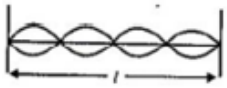
When pulse is reflected from a rigid support, the pulse is inverted both lengthwise and sidewise

Q.7 Answer:

$\frac{\sqrt{3}a}{2}$

Solution:

For a string vibrating in its nth **overtone** $= (n+1)^{th}$ **harmonic)**



$y = 2A\sin\left(\frac{(n+1)\pi x}{L}\right)\cos\omega t$

For $x = \frac{1}{3}, 2A = a$ **and** $n = 3$

$$y = \left[a\sin\left(\frac{4\pi}{l}\times\frac{l}{3}\right)\right]\cos\omega t$$

$$= a\sin\frac{4\pi}{3}\cos\omega t = -a\left(\frac{\sqrt{3}}{2}\right)\cos\omega t$$

Q.8 Answer:

9

Solution:

The standing wave pattern in the figure corresponds to $n = 3$ **(the** 3rd **harmonic) for a tube open at only one end.**
The length of the tube is $L = \frac{nv}{4f_n} = \frac{(3)(343\text{ m/s})}{4(660\text{ Hz})} = 0.39\text{ m}.$
$\Rightarrow 39\text{ cm} \Rightarrow (30+9)$. **So,** $x = 9$

Q.9 Answer:

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$a \sin \frac{\pi x}{L} \sin \frac{\pi y}{L}$

Solution:

Since the edges are clamped, displacement of the edges u $(x, y) = 0$ for line OA i.e. $\mathrm{y} = 0, 0 \leq \mathrm{x} \leq \mathrm{L}$

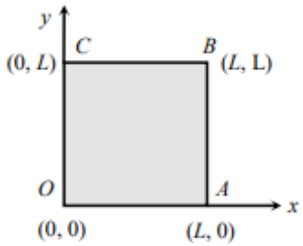
AB i.e. $x = L, 0 \leq y \leq L$

BC i.e. $\mathrm{y} = \mathrm{L}, \quad 0 \leq \mathrm{x} \leq \mathrm{L}$

OC i.e. $x = 0, 0 \leq y \leq L$

The above conditions are satisfied only in alternatives (B) and (C).

Note that $u(x, y) = 0$, for all four values e.g. in alternative $(D), u(x, y) = 0$ for $y = 0, y = L$ but it is not zero for $x = 0$ or $x = L$. Similarly in option $(A). u(x, y) = 0$ at $x = L, y = L$ but is not zero for $\mathrm{x} = 0$ or $\mathrm{y} = 0$, while in options (B) and $(\mathrm{C}), \mathrm{u}(\mathrm{x}, \mathrm{y}) = 0$ for $\mathrm{x} = 0, \mathrm{y} = 0, \mathrm{x} = \mathrm{L} \& \mathrm{y} = \mathrm{L}$.



Q.10 Answer:

It is greater at higher temperatures, but only in the case of a moving source and a stationary observer.

Solution:

In order for the Doppler effect to be large, the speed v_s of the source and/or the speed v_0 of the observer must be appreciable fractions of the speed of sound.

The Doppler effect depends on $\mathrm{v_s}/\mathrm{v}$ or $\mathrm{v_o}/\mathrm{v}$ or on both of these ratios.

For given values of $\mathrm{v_S}$ and $\mathrm{v_o}$, these ratios decrease, and the Doppler effect decreases as the speed of sound increases.

The speed of sound in air (assumed to be an ideal gas) increases with temperature, therefore, the Doppler effect decreases with increasing temperature, no matter if the source moves, the observer moves, or both move.

Q.11 Answer:

1

Solution:

If we add pulses 1 and 4 as per the principle of linear superposition, the resultant is a straight horizontal line that extends across the entire graph.

Q.12 Answer:

At the closed end of the air column, there is a displacement node and a pressure antinode

Solution:

At closed end displacement node and pressure antinode is formed.

$\frac{\lambda}{4} = 10 \text{ cm.}, \mathrm{v} = \mathrm{f}\lambda \quad ; \mathrm{f} = \frac{344}{40 \times 10^{-2}} = 860 \text{ Hz}$

For second resonance, $\ell = \frac{3\lambda}{4} = 30 \text{ cm.}$

Q.13 Answer:

860 Hz

Solution:

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At closed end displacement node and pressure antinode is formed.

$\frac{\lambda}{4} = 10 \text{ cm.}, \text{v} = \text{f}\lambda \quad ; \text{f} = \frac{344}{40 \times 10^{-2}} = 860 \text{ Hz}$

For second resonance, $\ell = \frac{3\lambda}{4} = 30 \text{ cm.}$

Q.14 Answer:

30 cm

Solution:

At closed end displacement node and pressure antinode is formed.

$\frac{\lambda}{4} = 10 \text{ cm.}, \text{v} = \text{f}\lambda \quad ; \text{f} = \frac{344}{40 \times 10^{-2}} = 860 \text{ Hz}$

For second resonance, $\ell = \frac{3\lambda}{4} = 30 \text{ cm.}$

Q.15 Answer:

C

Solution:

Coating tuning fork A will decrease its frequency, beat frequency increases means

$$\text{Given } \text{f}_\text{B} > \text{f}_\text{A} \ \& \ \text{f}_\text{C} > \text{f}_\text{A} \Rightarrow (\text{f}_\text{B} - \text{f}_\text{A}) < (\text{f}_\text{C} - \text{f}_\text{A}) \Rightarrow \ \text{f}_\text{C} > \text{f}_\text{B} > \text{f}_\text{A}$$

Q.16 Answer:

B

Solution:

Coating tuning fork A will decrease its frequency, beat frequency increases means

$\text{f}_\text{B} > \text{f}_\text{A} \& \text{f}_\text{C} > \text{f}_\text{A}$

$\text{Given } (\text{f}_\text{B} - \text{f}_\text{A}) < (\text{f}_\text{C} - \text{f}_\text{A})$

$\text{f}_\text{C} > \text{f}_\text{B} > \text{f}_\text{A}$

Q.17 Answer:

Both (A) and (B)

Solution:

Coating tuning fork A will decrease its frequency, beat frequency increases means

$\text{f}_\text{B} > \text{f}_\text{A} \& \text{f}_\text{C} > \text{f}_\text{A}$

$\text{Given } (\text{f}_\text{B} - \text{f}_\text{A}) < (\text{f}_\text{C} - \text{f}_\text{A})$

$\text{f}_\text{C} > \text{f}_\text{B} > \text{f}_\text{A}$

Q.18 Answer:

A-(pqs) B-(qr) C-(qr) D-(pqs)

Solution:

$(A) - pqs, (B) - qr, (C) - qr, (D) - pqs$

In open organ pipe all harmonics are possible.

In closed organ pipe odd harmonics are possible.

Q.19 Answer:

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Solution:

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By using $\mathrm{n'=n\left(\frac{v}{v-v_s}\right)}$
$\Rightarrow \mathrm{f_1=n\left(\frac{v}{v-v_s}\right)=n\left(\frac{340}{340-34}\right)=\frac{340}{306}n}$ **and** $\mathrm{f_2=n\left(\frac{340}{340-17}\right)=\frac{340}{323}n;\frac{f_1}{f_2}=\frac{323}{306}=\frac{19}{18}}$
$=\frac{10+9}{18}$ **So** $n=9$

Q.20 Answer:

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Solution:

IDENTIFY: Apply $f_{\mathrm{L}}=\left(\frac{v+v_{\mathrm{L}}}{v+v_{\mathrm{S}}}\right) f_{\mathrm{S}}$**. The heart wall first acts as the listener and then as the source.**

SET UP: The positive direction is from listener to source. The heart wall is moving toward the receiver so the Doppler effect increases the frequency and the final frequency received, f_{L2}**, is greater than the source frequency,** $f_{\mathrm{S1}} \cdot f_{\mathrm{L2}}-f_{\mathrm{S1}}=72 \mathrm{~Hz}$.

EXECUTE: Heart wall receives the sound: $f_{\mathrm{S}}=f_{\mathrm{S1}}, \quad f_{\mathrm{L}}=f_{\mathrm{L1}}, \quad v_{\mathrm{S}}=0, \quad v_{\mathrm{L}}=-v_{\text{wall}} \cdot \quad f_{\mathrm{L}}=\left(\frac{v+v_{\mathrm{L}}}{v+v_{\mathrm{S}}}\right) f_{\mathrm{S}}$ **gives** $f_{\mathrm{L1}}=\left(\frac{v-v_{\text{wall}}}{v}\right) f_{\mathrm{Sl}}$

Heart wall emits the sound: $f_{\mathrm{S2}}=f_{\mathrm{L1}^*} \cdot v_{\mathrm{S}}=+v_{\text{wall}} \cdot v_{\mathrm{L}}=0$.

$$f_{\mathrm{L2}}=\left(\frac{v}{v+v_{\text{wall}}}\right) f_{\mathrm{S2}}=\left(\frac{v}{v+v_{\text{wall}}}\right)\left(\frac{v-v_{\text{wall}}}{v}\right) f_{\mathrm{S1}}=\left(\frac{v-v_{\text{wall}}}{v+v_{\text{wall}}}\right) f_{\mathrm{S1}^*}.$$

$$f_{\mathrm{L2}}-f_{\mathrm{SI}}=\left(1-\frac{v-v_{\text{wall}}}{v+v_{\text{wall}}}\right) f_{\mathrm{S1}}=\left(\frac{2v_{\text{wall}}}{v+v_{\text{wall}}}\right) f_{\mathrm{S1}}-v_{\text{wall}}=\frac{\left(f_{\mathrm{L2}}-f_{\mathrm{S1}}\right) v}{2f_{\mathrm{S1}}-\left(f_{\mathrm{L2}}-f_{\mathrm{S1}}\right)} \cdot f_{\mathrm{S1}}\mathrm{Jo}f_{\mathrm{L2}}-f_{\mathrm{S1}} \text{ and}$$

$$v_{\text{wall}}=\frac{\left(f_{\mathrm{L2}}-f_{\mathrm{S1}}\right) v}{2f_{\mathrm{S1}}}=\frac{(72 \mathrm{~Hz})(1500 \mathrm{~m/s})}{2\left(2.00 \times 10^6 \mathrm{~Hz}\right)}=0.0270 \mathrm{~m/s}=2.70 \mathrm{~cm/s}$$

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